

Program : Diploma in Civil and Environmental Engineering	
Course Code: 5352B	Course Title: Environmental Impact Assessment
Semester: 5	Credits: 3
Course Category: Programme Elective Course	
Periods per week: 3 (L:3 T:0 P:0)	Periods per semester: 45

Course Objectives:

- To provide an insight into the concept of environmental impact assessment and its purpose.
- To enable students to identify various impacts associated with infrastructure projects.

Course Prerequisites:

Topic	Course code	Course name	Semester
Sustainable development	4353	Sustainable engineering	4
Water pollution and air pollution	4353	Sustainable engineering	4
Environmental management standards and life cycle analysis	4353	Sustainable engineering	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Demonstrate process, need, significance and methods of environmental impact analysis.	12	Understanding
CO2	Identify impact of various pollutants on quality of water and soil.	10	Understanding
CO3	Identify causes and impacts of air and noise pollution.	11	Understanding
CO4	Identify socio economic, ecological impacts of infrastructure projects and discuss case studies on EIA.	10	Understanding
	Tests	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3				3		
CO2	3				3		
CO3	3				3		
CO4	3				3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate process, need, significance and methods of environmental impact analysis.		
M1.01	Discuss concept, purpose and benefits of EIA	2	Understanding
M1.02	Identify key features of EIA notification 2020	2	Understanding
M1.03	Illustrate various steps in EIA process	4	Understanding
M1.04	Identify various methods of impact analysis	4	Understanding
	Total	12	
Contents: Concept of environmental impact analysis History of EIA - Global and Indian scenario, Need for EIA, Benefits of EIA. Key features of EIA notification 2020. EIA Process in India, Public participation – Significance & steps, Environment management plan. Impact analysis- Adhoc, checklists, matrix methods, overlay analysis, Fault Tree Analysis method & Event Tree Analysis method Role of an Environmental Engineer			
CO2	Identify impact of various pollutants on quality of water and soil.		

M2.01	Identify different types of environmental impacts	2	Understanding
M2.02	Identify various sources of water pollution and impacts of water pollution.	4	Understanding
M2.03	Identify characteristics of various types of solid waste and its impact on environment.	4	Understanding
	Series 1	1	Understanding
	Total	11	

Contents

Types of impact- biophysical, social, health and social, Direct, indirect and cumulative, Positive and negative, random and predictable, local and widespread, short and long term.

Impacts of Water Pollution-Point and Non-point Source of Pollution, Major Pollutants of Water, Physical, chemical and biological characteristics of water, Projects that create concerns for surface water environment, mitigation of impacts.

Impacts of Solid Waste-Classification and sources of Solid Waste, Characteristics of Solid Waste, E-waste, & Radioactive wastes - Types, management/disposal, Hazardous waste - waste identification process and characteristics.

Solid Waste Management Rules 2016

CO3	Identify causes and impacts of air and noise pollution.		
M3.01	Identify impacts of urbanization and modern agricultural practices on land and its mitigation.	4	Understanding
M3.02	Identify causes and impacts of air pollution	4	Understanding
M3.03	Identify causes and impacts of noise pollution	3	Understanding
	Total	11	

Contents:

Land/Soil Pollution- Effects of urbanization on land degradation, Impact of Modern Agriculture on Soil, pesticide pollution, Effect on Environment, mitigation of impacts.

Impacts of Air Pollution-Classification of Pollution and Pollutants, Primary and Secondary Pollutants, Criteria Pollutants and their impacts on environment, human health, National Ambient Air Quality Standards by CPCB, mitigation of impacts.

Noise Pollution-Sources of Noise, Effects of Noise, measurement of noise, Equivalent sound

pressure level, Control measures –Noise pollution (Regulation and control) Rule 2000			
CO4	Identify socio economic, ecological impacts of infrastructure projects and discuss case studies on EIA.		
M4.01	Identify socio economic impacts of infrastructure projects.	3	Understanding
M4.02	Identify impact of infrastructure projects on flora and fauna.	2	Understanding
M4.03	Conduct case studies on EIA of infrastructure projects.	5	Understanding
	Series 2	1	
	Total	11	
Contents: Socio economic impact Analysis- Definition, types of socio-economic impacts, socio economic factors and potential changes resulting from project implementation-examples. Steps in conducting social impact assessment, principles of social impact assessment. Ecological impact – Impact of infrastructure projects on flora and fauna. EIA case studies for infrastructure products-Highways, multi storied buildings, dams, water supply projects, ports – Any 3 case studies.			

Text / Reference:

T/R	Book Title/Author
T	John Glasson, Riki Therivel & S Andrew Chadwick, Introduction to EIA University College London Press Limited, 2005
T	Larry W Canter, Environmental Impact Assessment, McGraw Hill Inc., New York 1996
T	Y Anjaneyalu, ValliManickam, Environmental Impact Assessment Methodologies, B S Publications, 2007
R	MINISTRY OF ENVIRONMENT, FOREST AND CLIMATE CHANGE, EIA notification 2020

Online resources:

Sl.No.	Website Link
1	https://youtube.com/playlist?list=PLLy_2iUCG87CkrNdXME16BCptwGx1fl67&si=k5qPNUfOsAHw5Hne